

Step 1: Create the Database

The screenshot shows the MySQL Workbench interface. The left sidebar contains the 'MANAGEMENT' section with options like Server Status, Client Connections, and Users and Privileges. The 'INSTANCE' section shows Startup / Shutdown, Server Logs, and Options File. The 'PERFORMANCE' section shows Dashboard, Performance Reports, and Performance Schema Setup. The 'Administration' tab is selected, showing 'Schemas' and 'No object selected'.

The main editor window shows a SQL script with the following queries:

```
1 CREATE DATABASE student_system;
2 USE student_system;
3
4 show databases;
```

The 'Result Grid' shows the output of the queries:

Database
information_schema
mysql
performance_schema
student_system
sys

The 'Output' window shows the 'Action Output' table:

#	Time	Action	Message	Duration / Fetch
1	09:52:32	CREATE DATABASE student_system	1 row(s) affected	0.015 sec
2	09:52:32	USE student_system	0 row(s) affected	0.000 sec
3	09:52:32	show databases	5 row(s) returned	0.000 sec / 0.000 sec

Step2: Create Core Table Students Table and insert data

The screenshot shows the MySQL Workbench interface. The left sidebar is the same as in Step 1. The 'Administration' tab is selected, showing 'Schemas' and 'No object selected'.

The main editor window shows a SQL script with the following queries:

```
1 CREATE TABLE Students (
2     StudentID INT AUTO_INCREMENT PRIMARY KEY,
3     Name VARCHAR(100),
4     Email VARCHAR(100) UNIQUE,
5     Address VARCHAR(255),
6     Country VARCHAR(50),
7     Course VARCHAR(100)
8 );
9
10 INSERT INTO Students (Name, Email, Address, Country, Course) VALUES
11 ('Anjali Mehta', 'anjali@example.com', 'Mumbai, India', 'India', 'Computer Science'),
12 ('John Smith', 'john.smith@example.com', 'New York, USA', 'USA', 'Mathematics');
```

The 'Result Grid' shows the output of the queries:

StudentID	Name	Email	Address	Country	Course
1	Anjali Mehta	anjali@example.com	Mumbai, India	India	Computer Science
2	John Smith	john.smith@example.com	New York, USA	USA	Mathematics
3	Ravi Kumar	ravi.k@example.com	Delhi, India	India	Physics
4	Emily Johnson	emily.j@example.com	Los Angeles, USA	USA	Biology

The 'Output' window shows the 'Action Output' table:

#	Time	Action	Message	Duration / Fetch
1	09:55:50	CREATE TABLE Students (StudentID INT AUTO_INCREMENT PRIMARY KEY, Name ...	0 row(s) affected	0.047 sec
2	09:55:50	INSERT INTO Students (Name, Email, Address, Country, Course) VALUES ('Anjali Mehta', 'anj...	4 row(s) affected Records: 4 Duplicates: 0 Warnings: 0	0.000 sec
3	09:55:50	select * from Students LIMIT 0, 1000	4 row(s) returned	0.000 sec / 0.000 sec

Step 3: Create Fragmented Tables and Insert Data into Fragments (HORIZONTAL FRAGMENTATION)

The screenshot shows the MySQL Workbench interface. The central SQL editor contains the following SQL code:

```
1 CREATE TABLE Students_India (  
2     StudentID INT PRIMARY KEY,  
3     Name VARCHAR(100),  
4     Email VARCHAR(100) UNIQUE,  
5     Address VARCHAR(255),  
6     Country VARCHAR(50),  
7     Course VARCHAR(100)  
8 );  
9  
10 CREATE TABLE Students_USA (  
11     StudentID INT PRIMARY KEY,  
12     Name VARCHAR(100),  
13     Email VARCHAR(100) UNIQUE,  
14     Address VARCHAR(255),  
15     Country VARCHAR(50),  
16     Course VARCHAR(100)  
17 );  
18 INSERT INTO Students_India (StudentID, Name, Email, Address, Country, Course)  
19 SELECT StudentID, Name, Email, Address, Country, Course
```

The Output pane at the bottom shows the execution results:

#	Time	Action	Message	Duration / Fetch
2	09:58:43	CREATE TABLE Students_USA (StudentID INT PRIMARY KEY, Name VARCHAR(100),	0 row(s) affected	0.031 sec
3	09:58:43	INSERT INTO Students_India (StudentID, Name, Email, Address, Country, Course) SELECT ...	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0	0.000 sec
4	09:58:43	INSERT INTO Students_USA (StudentID, Name, Email, Address, Country, Course) SELECT ...	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0	0.016 sec

Step 4: Querying Data

The screenshot shows the MySQL Workbench interface. The central SQL editor contains the following SQL code:

```
1 -- Indian students  
2 SELECT * FROM Students_India;  
3  
4 -- USA students  
5 SELECT * FROM Students_USA;  
6  
7 -- Combined  
8 SELECT * FROM Students_India  
9 UNION ALL  
10 SELECT * FROM Students_USA;  
11
```

The Output pane at the bottom shows the execution results:

#	Time	Action	Message	Duration / Fetch
1	09:59:51	SELECT * FROM Students_India LIMIT 0,1000	2 row(s) returned	0.000 sec / 0.000 sec
2	09:59:51	SELECT * FROM Students_USA LIMIT 0,1000	2 row(s) returned	0.000 sec / 0.000 sec
3	09:59:51	SELECT * FROM Students_India UNION ALL SELECT * FROM Students_USA	4 row(s) returned	0.000 sec / 0.000 sec

Step 5: Create Fragmented Tables and Insert Data into Fragments (VERTICAL FRAGMENTATION)

The screenshot shows the MySQL Workbench interface. The left sidebar contains the 'MANAGEMENT' section with options like Server Status, Client Connections, Users and Privileges, Status and System Variables, Data Export, and Data Import/Restore. The 'INSTANCE' section includes Startup / Shutdown, Server Logs, and Options File. The 'PERFORMANCE' section includes Dashboard, Performance Reports, and Performance Schema Setup. The 'Administration' tab is selected, showing 'Schemas' and 'Information' sections. The 'Information' section shows 'No object selected'.

The main editor window displays the following SQL script:

```
5 Password VARCHAR(255)
6 );
7
8 -- Profile Info
9 CREATE TABLE Student_Profile (
10 StudentID INT PRIMARY KEY,
11 Name VARCHAR(100),
12 Address VARCHAR(255),
13 Country VARCHAR(50),
14 Course VARCHAR(100)
15 );
16
17 -- Add fake passwords for now (in real use, hash them)
18 INSERT INTO Student_Auth (StudentID, Email, Password)
19 SELECT StudentID, Email, 'fake_hashed_password' FROM Students;
20
21 INSERT INTO Student_Profile (StudentID, Name, Address, Country, Course)
22 SELECT StudentID, Name, Address, Country, Course FROM Students;
23
```

The right sidebar shows 'SQL Additions' with a message: 'Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.'

The 'Output' window shows the following results:

#	Time	Action	Message	Duration / Fetch
2	10:04:13	CREATE TABLE Student_Profile (StudentID INT PRIMARY KEY, Name VARCHAR(100)...	0 row(s) affected	0.031 sec
3	10:04:13	INSERT INTO Student_Auth (StudentID, Email, Password) SELECT StudentID, Email, 'fake_...	4 row(s) affected Records: 4 Duplicates: 0 Warnings: 0	0.000 sec
4	10:04:13	INSERT INTO Student_Profile (StudentID, Name, Address, Country, Course) SELECT Studen...	4 row(s) affected Records: 4 Duplicates: 0 Warnings: 0	0.016 sec

Step 6: Join and View Combined Data

The screenshot shows the MySQL Workbench interface. The left sidebar is the same as in Step 5. The 'Administration' tab is selected, showing 'Schemas' and 'Information' sections. The 'Information' section shows 'No object selected'.

The main editor window displays the following SQL script:

```
1 SELECT sa.StudentID, sa.Email, sp.Name, sp.Address, sp.Country, sp.Course
2 FROM Student_Auth sa
3 JOIN Student_Profile sp ON sa.StudentID = sp.StudentID;
4
```

The right sidebar shows 'SQL Additions' with a message: 'Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.'

The 'Output' window shows the following results:

StudentID	Email	Name	Address	Country	Course
1	anjali@example.com	Anjali Mehta	Mumbai, India	India	Computer Science
4	emily.j@example.com	Emily Johnson	Los Angeles, USA	USA	Biology
2	john.smith@example.com	John Smith	New York, USA	USA	Mathematics
3	ravi.k@example.com	Ravi Kumar	Delhi, India	India	Physics

The 'Output' window also shows the following results:

#	Time	Action	Message	Duration / Fetch
1	10:04:13	CREATE TABLE Student_Auth (StudentID INT PRIMARY KEY, Email VARCHAR(100)...	0 row(s) affected	0.031 sec
2	10:04:13	CREATE TABLE Student_Profile (StudentID INT PRIMARY KEY, Name VARCHAR(100)...	0 row(s) affected	0.031 sec
3	10:04:13	INSERT INTO Student_Auth (StudentID, Email, Password) SELECT StudentID, Email, 'fake_...	4 row(s) affected Records: 4 Duplicates: 0 Warnings: 0	0.000 sec
4	10:04:13	INSERT INTO Student_Profile (StudentID, Name, Address, Country, Course) SELECT Studen...	4 row(s) affected Records: 4 Duplicates: 0 Warnings: 0	0.016 sec
5	10:05:46	SELECT sa.StudentID, sa.Email, sp.Name, sp.Address, sp.Country, sp.Course FROM Student_...	4 row(s) returned	0.000 sec / 0.000 sec